

AI Usage DISCLOSURE FORM

Samsung PRISM Generative AI Hackathon — 3rd Edition (2026–27)

Theme #03: Teachable Voice Automation

1. Team Details

Team Name: Cheesecake
Project Name: FlowPilot — Learn-Once, Replay-Anywhere Voice Automation
Organization: SRM Institute of Science and Technology (SRMIST), Kattankulathur
Date: September 30, 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? YES [X] / NO []

Summary: Generative AI and NLP models were used for dynamic voice intent parsing, parameter slot generalization from raw touch traces, and speech recognition.

3. Purpose of AI Usage (Brief Details)

- **Idea generation:** Evaluated Accessibility Tree abstractions and parameter slot generalization strategies for Android voice automation.
- **Code assistance:** Assisted in drafting boilerplate for Retrofit HTTP clients, FastAPI route scaffolds, and benchmark test harnesses.
- **UI / UX design:** Assisted in selecting Material 3 color tokens and Jetpack Compose component styling.
- **Content creation:** Architecture diagrams, sequence flows, and comprehensive API documentation specifications.
- **Data analysis:** Benchmark evaluation metrics calculation across 25 queries, 5 target apps, and latency distribution analysis.
- **Testing / debugging:** Automated unit tests and test client assertions for vector matching and slot compilation.

4. Feature Origin Classification

• Feature 1: Generalized Flow Compilation (LLM Flow Synthesis) [Classification: Hybrid]

AI Platform: Google Gemini 2.0 Flash / Gemini 1.5 Flash

Description: Raw UI action traces recorded by AccessibilityService are fed into Gemini with structured prompts to identify dynamic parameters (search queries, quantities) and emit an abstract JSON FlowGraph.

Modifications: Added strict Pydantic validation, fallback deterministic heuristic compilers, and token truncation filters.

• Feature 2: Voice Command Semantic Intent Matching [Classification: Hybrid]

AI Platform: Sentence-BERT (all-MiniLM-L6-v2) + ChromaDB

Description: Natural language voice commands are projected into a 384-dimensional dense semantic embedding space with cosine similarity against stored flows.

Modifications: Confidence score thresholding (0.75 cutoff) to trigger negative intent rejection (T12) and ambiguity clarification (T13).

• Feature 3: Voice Speech-to-Text [Classification: AI-Integrated]

AI Platform: OpenAI Whisper (Backend service) & Android Native SpeechRecognizer.

Modifications: Configured phonetic biasing and ambient noise thresholding for Indian English accents.

• Feature 4: Android Accessibility Execution Engine [Classification: 100% Self-Generated / Student Engineered]

Description: Custom Kotlin code leveraging Android AccessibilityService. Implements a 4-tier cascading element finder (resource-id -> content-desc -> text -> bounding box), dynamic parameter injection, gesture dispatch, and security pauses for payment/OTP screens.

Modifications: Completely custom engineering without pre-existing commercial RPA SDKs.

5. Ethical & Compliance Confirmation

- AI usage complies with guidelines and policies: **YES [X]**
- No proprietary or copyrighted data misused: **I AGREE [X]**
- Security & Zero Credential Capture: FlowPilot explicitly enforces a hard boundary halting at payment, PIN, OTP, and password screens (T11). No sensitive user data is processed by LLMs.

6. Declaration & Sign-Off

Name of Team Representative: Parth Jain | **Role:** Team Lead / Primary Developer

Signature: Parth Jain (Submitted Electronically) | **Date:** September 30, 2026